

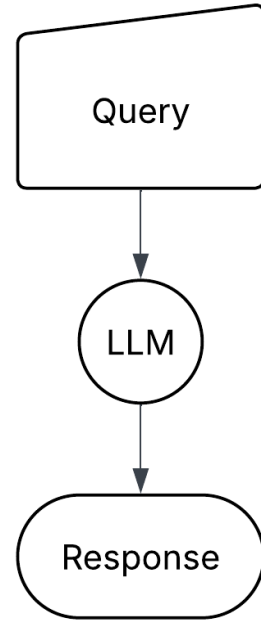
IMPERIAL

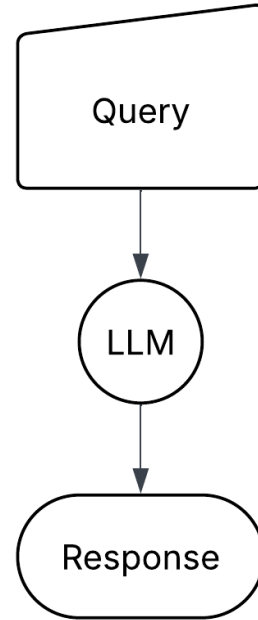
MultiRAG: A Framework to Fuse Expert Retrievers via Hybrid Reranking

Author: **Thomas Grant**

Supervisor: **Dr Nihir Vedd**

Second Marker: **Dr Chiraag Lala**



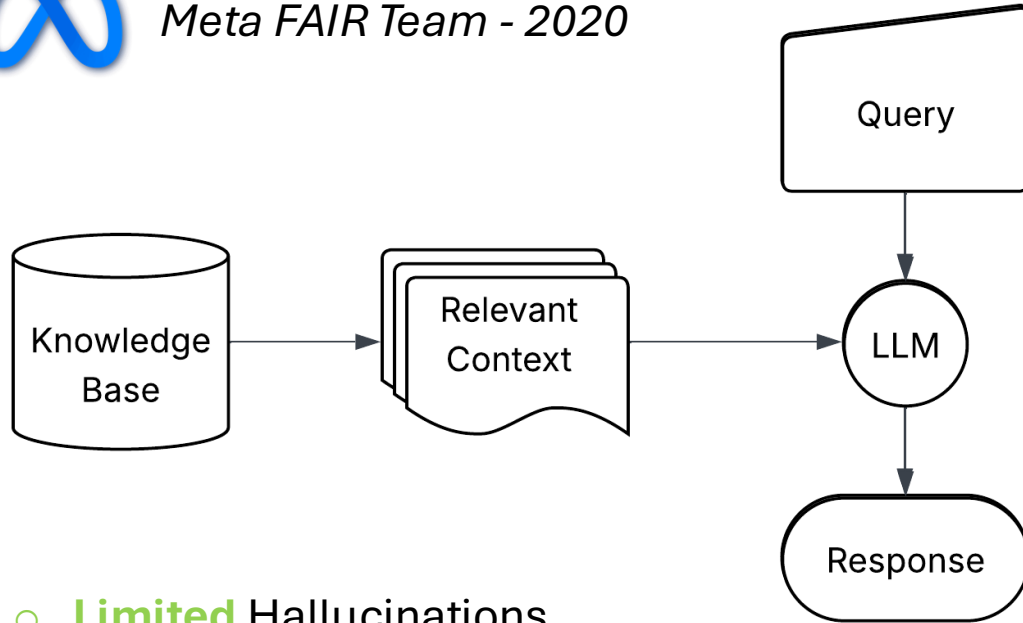


- Hallucinations
- Outdated knowledge
- Lack of Domain Specificity
- Static knowledge base
- Context length limits
- Inability to provide citations or provenance

Rise of RAG



Meta FAIR Team - 2020

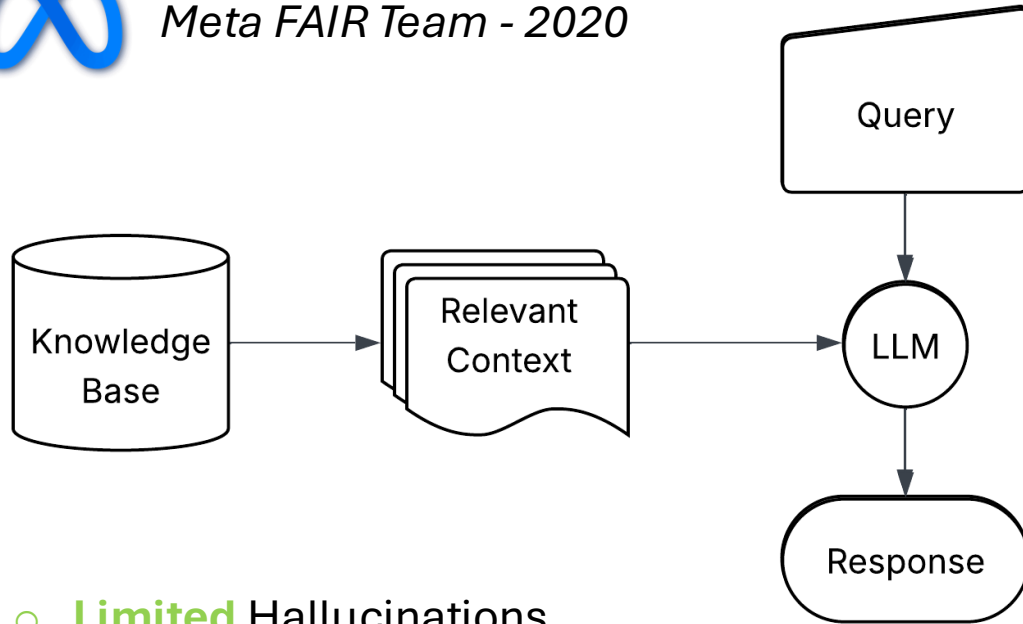


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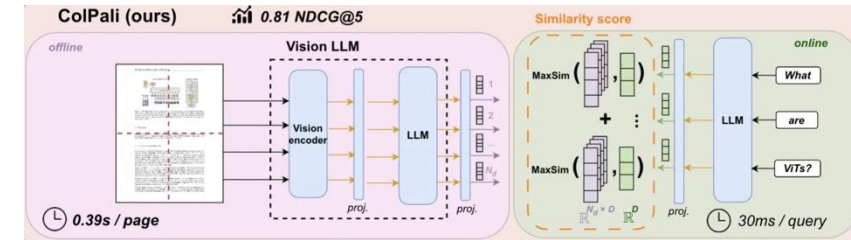


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Current SOTA systems

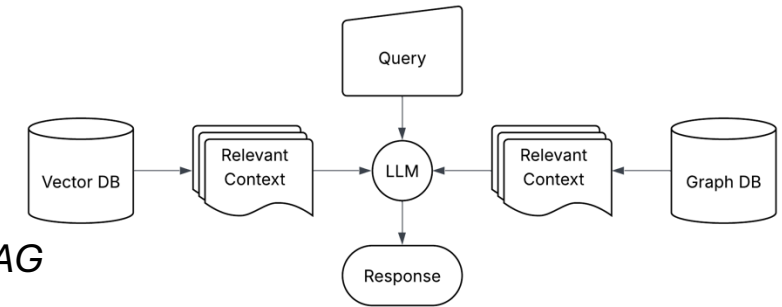
ColPali- 2024

Multimodal model
with ColBERT Late
Interaction (MaxSim)

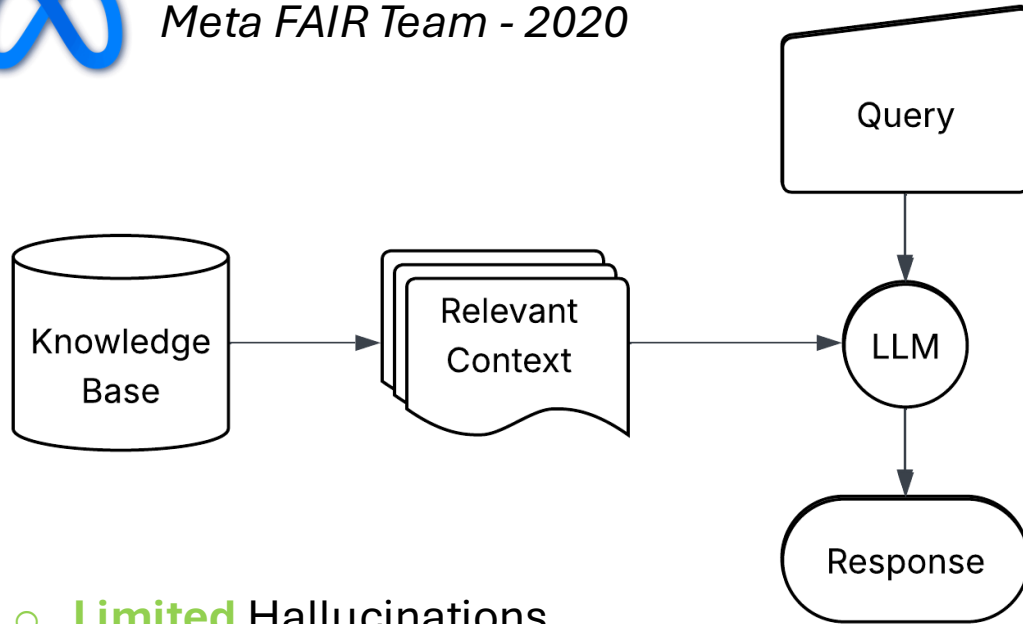

<https://huggingface.co/blog/manu/colpali>

HybridRAG – 2024

Concatenate the
contexts from
VectorRAG and GraphRAG



Rise of RAG

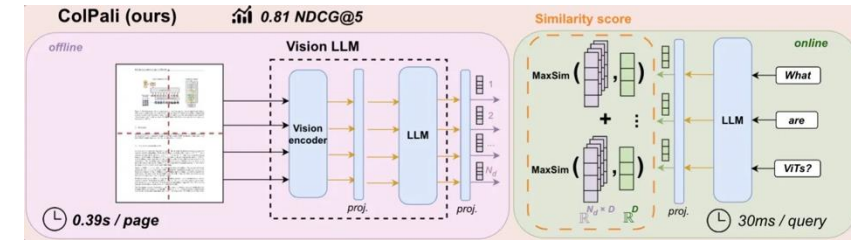


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Monolithic Limitation

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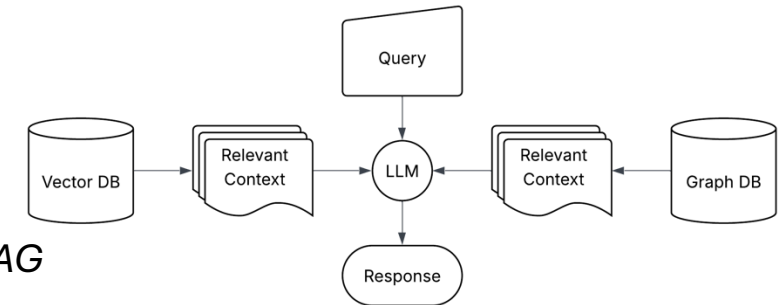
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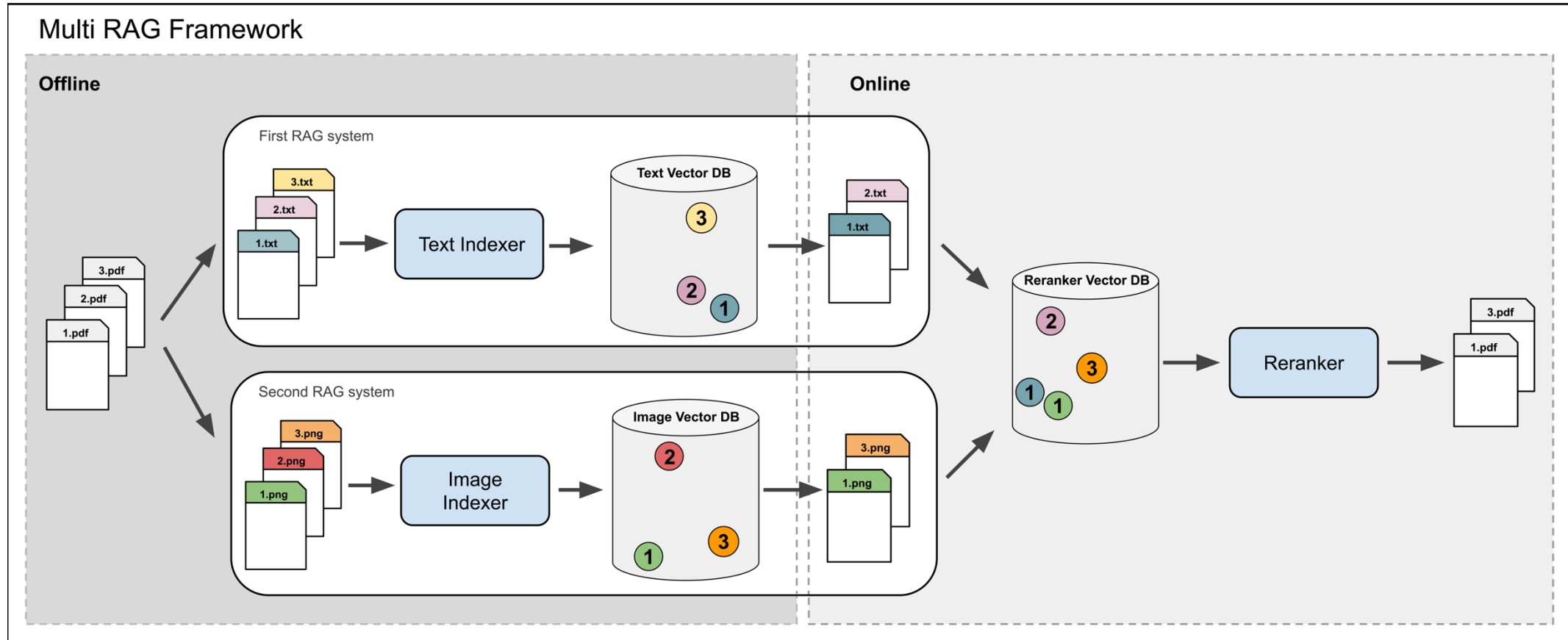
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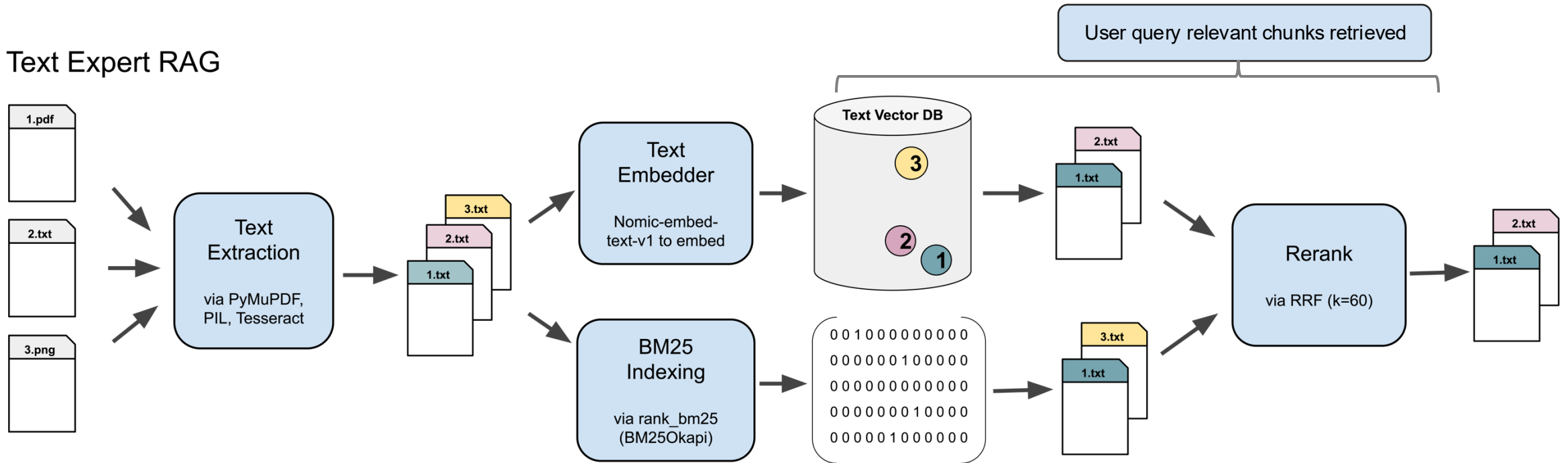
- ColPali excels with visual data
- HybridRAG excels with text-centric datasets

No single-retriever is optimal in all scenarios.

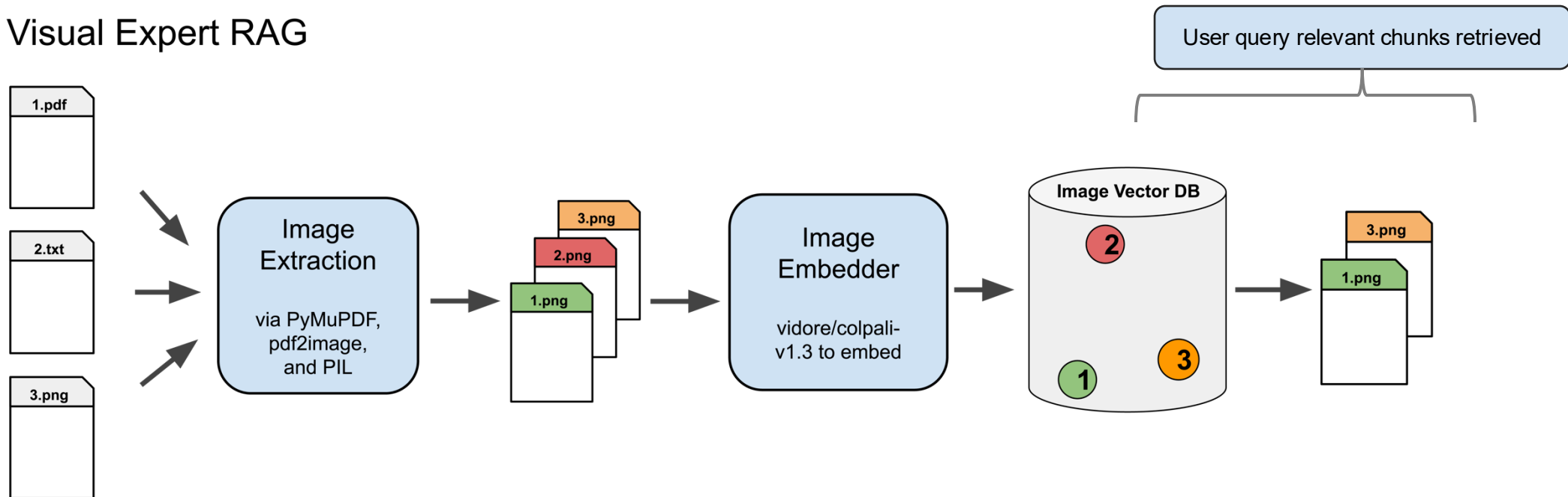


- Modular framework with multiple independent retrieval pipelines
- Each pipeline optimized for a specific data type
 - Text expert: Traditional RAG with hybrid (BM25 + semantic) retrieval
 - Visual expert: Multimodal RAG based on ColPali

Text Expert RAG



Visual Expert RAG



Hybrid Reranking



- More compute at inference
- Slow and expensive
- Context size limit

Hybrid Reranking



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- More compute during indexing
- More storage required

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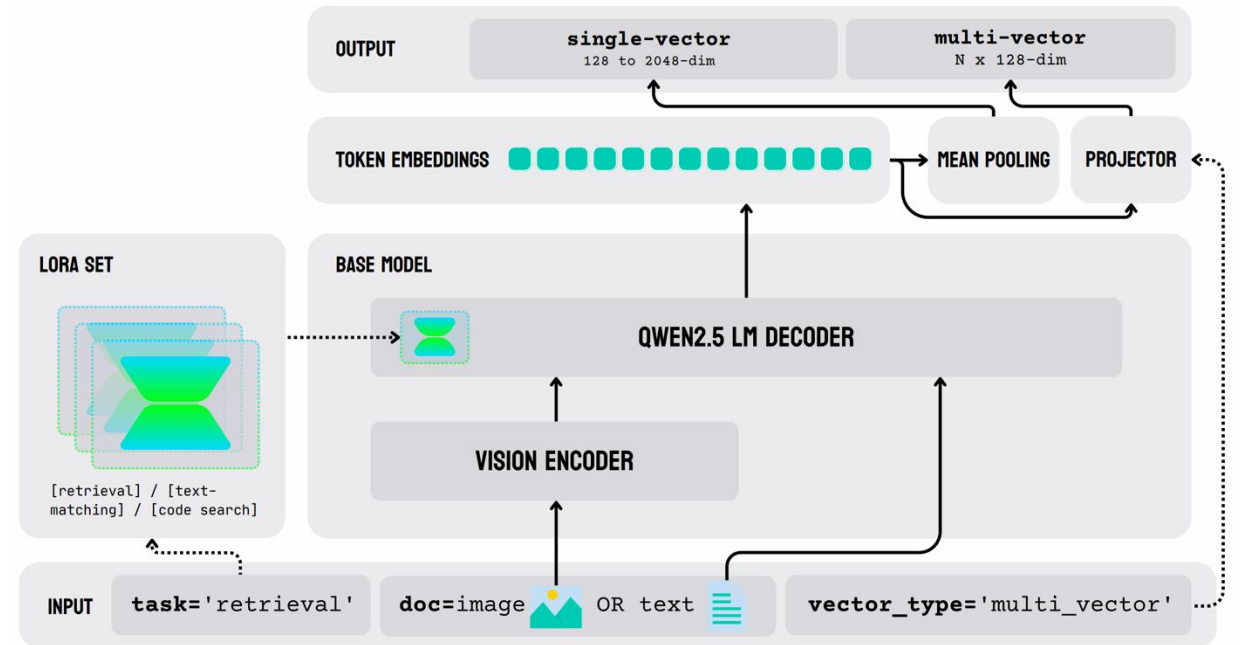


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Why Jina?



- Unlike CLIP-style models with separate encoders, Jina uses a shared encoder for text and images
- Open model

Fusion strategies

Reciprocal Rank Fusion (RFF) $\longrightarrow score = score + \frac{1}{k+rank_R^{(id)}}$

Text RAG	Id					
	2	3	1	5	7	4
	0.70	0.65	0.60	0.30	0.20	0.10

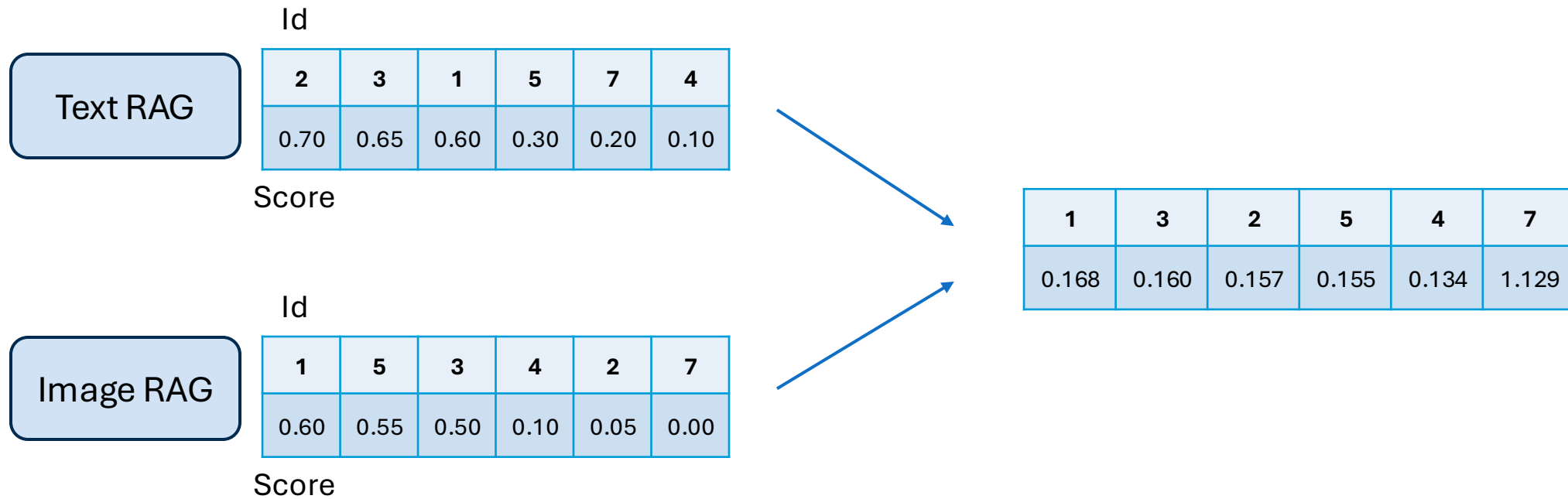
Score

Image RAG	Id					
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Score

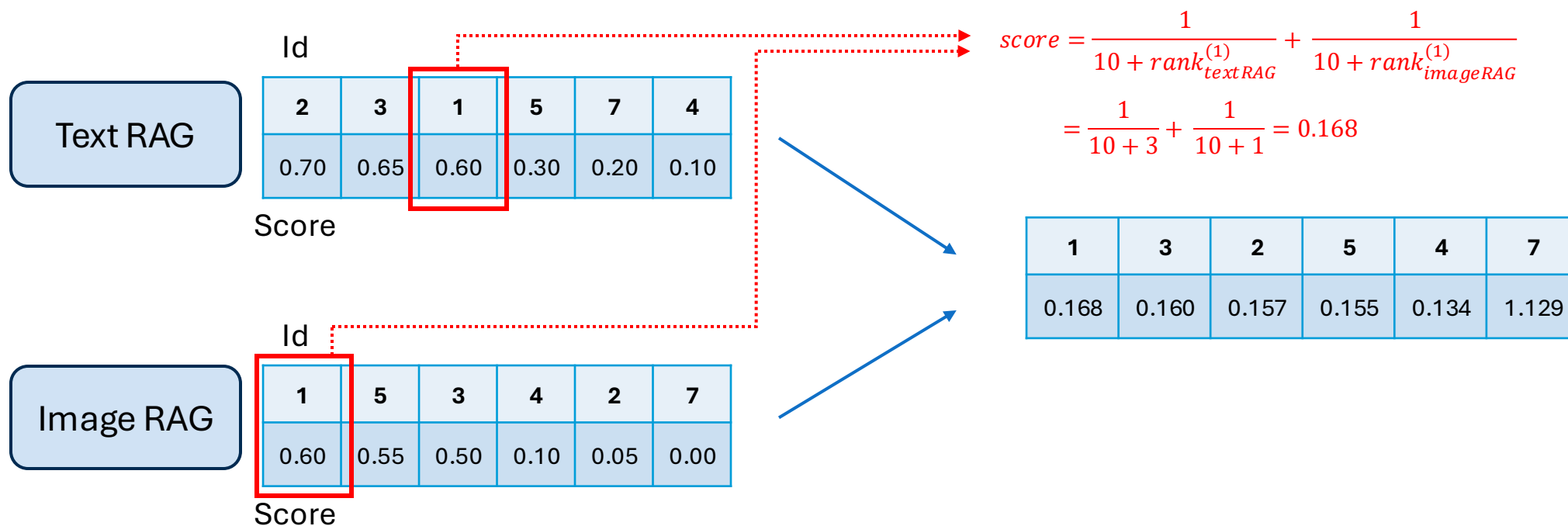
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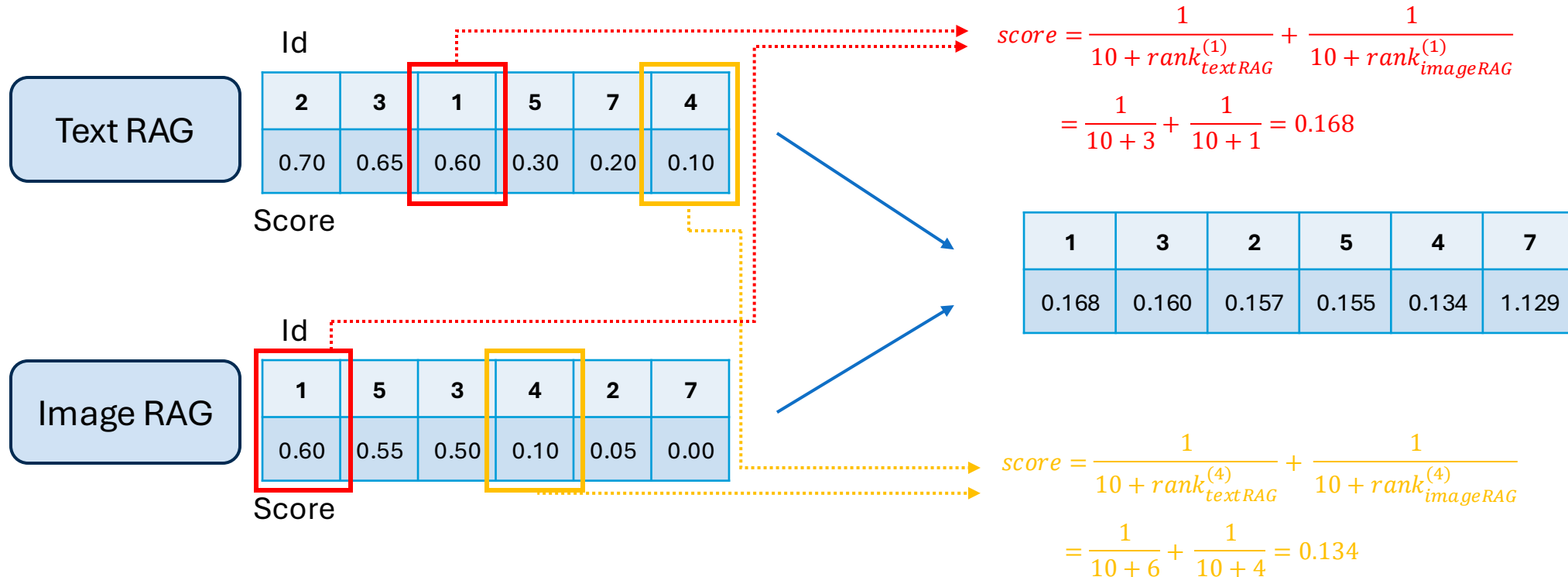
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Fusion strategies

Normalise, Average and Fuse

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L1 normalization

- Probabilistic interpretation, Good for proportional weighting

L2 normalization

- Smooth distribution of scores, good for ranking documents by closeness.

Why average?

- Favours documents that are found relevant across expert retrievers

Fusion strategies

Normalise, Average and Fuse

Min-max or Z-score may have been strong candidates as well

Seem to counter the “Mixture of Experts” approach

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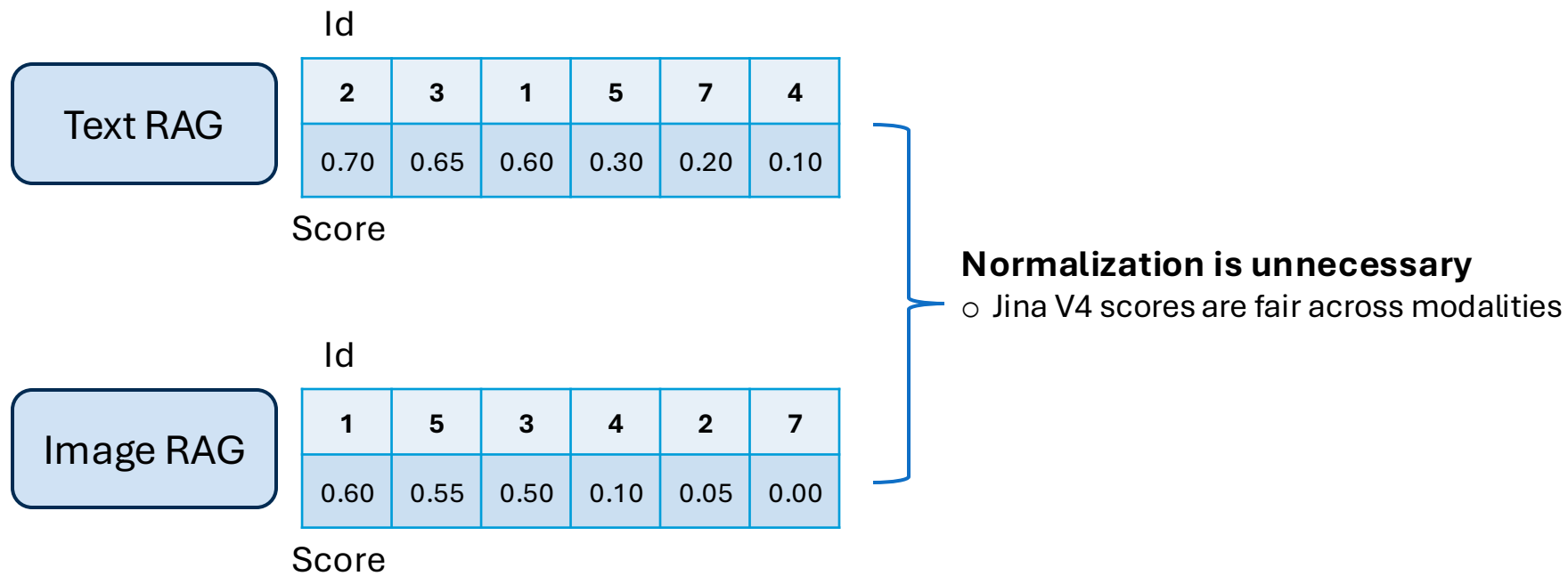
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Best rank across modalities

- Ensures to pick the best rank of each relevant chunks across its modalities.
- In-line with the "Mixture of Experts" approach

Fusion strategies

Simply Rerank and Fuse



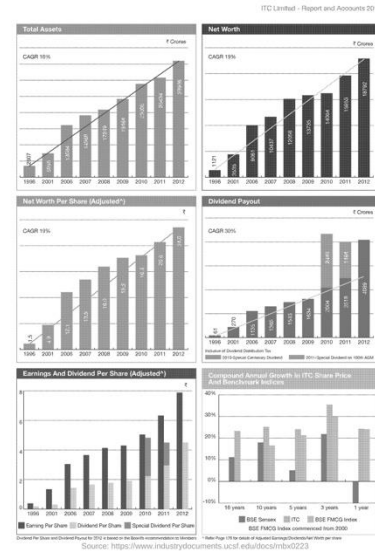
Existing benchmarks

"how many infrastructure in this diagram are new?"
(InfoVQA – query 13)



https://huggingface.co/datasets/vidore/infovqa_test_subsampled

"What is the dividend payout in 2012?"
(DocVQA – query 0)



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"What are the key risks identified in the global economic outlook?"
(EconomicsReportsV2 – query 1)

Key Takeaways

Debt servicing costs of low- and middle-income countries (LMICs), excluding China, reached an all-time high in 2023, double the level of a decade ago. Higher external debt levels—which also increased to an all-time high in 2023, to US\$8.8 trillion (including China)—coupled with elevated interest rates posed new and challenging debt burdens for these countries. Overall debt levels continued to rise as a result of accumulated borrowing during the pandemic and were exacerbated by financial pressures in its aftermath: notably, a sharp rise in global interest rates, depreciating local currencies, and uncertainty surrounding global economic growth.

In addition, the data indicate that the composition of LMICs' external borrowing has changed markedly since the onset of the pandemic, with multilateral creditors significantly increasing their share of lending to LMICs amid slower lending growth from private creditors.

Key takeaways from the 2023 data include the following:

- The total debt servicing costs (principal plus interest payments) of all LMICs reached an all-time high of US\$1.4 trillion in 2023. For LMICs, excluding China, debt servicing costs climbed to a record of US\$971.1 billion in 2023, an increase of 19.7 percent over the previous year and more than double the amounts seen a decade ago. These historically challenging debt service costs were due to record debt levels, interest rates at a two-decade high, and depreciation of local currencies against a strong US dollar.
- Since the onset of the pandemic, multilateral lenders have become the central financial lifeline for LMICs amid a slowing of private lending. The composition of LMICs' external debt portfolios has changed significantly since 2019 as multilateral creditors—including the International Monetary Fund, the World Bank, and regional development banks—stepped up and assumed the role of providers of emergency relief and balance of payments support in times of crisis. Borrowing from private creditors fell sharply because of adverse market conditions, investor retreat from frontier markets, and—in countries eligible for International Development Association (IDA) assistance—a concentration on borrowing from official creditors on concessional terms to support debt sustainability.
- Official multilateral creditors have been positive contributors to net transfers to LMICs throughout the past decade, offering concessional financing at low interest rates, long maturities, and support for countries through times

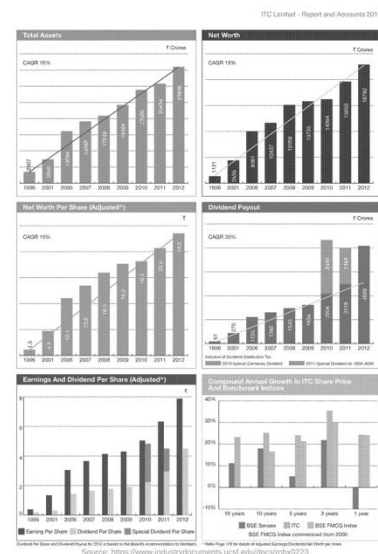
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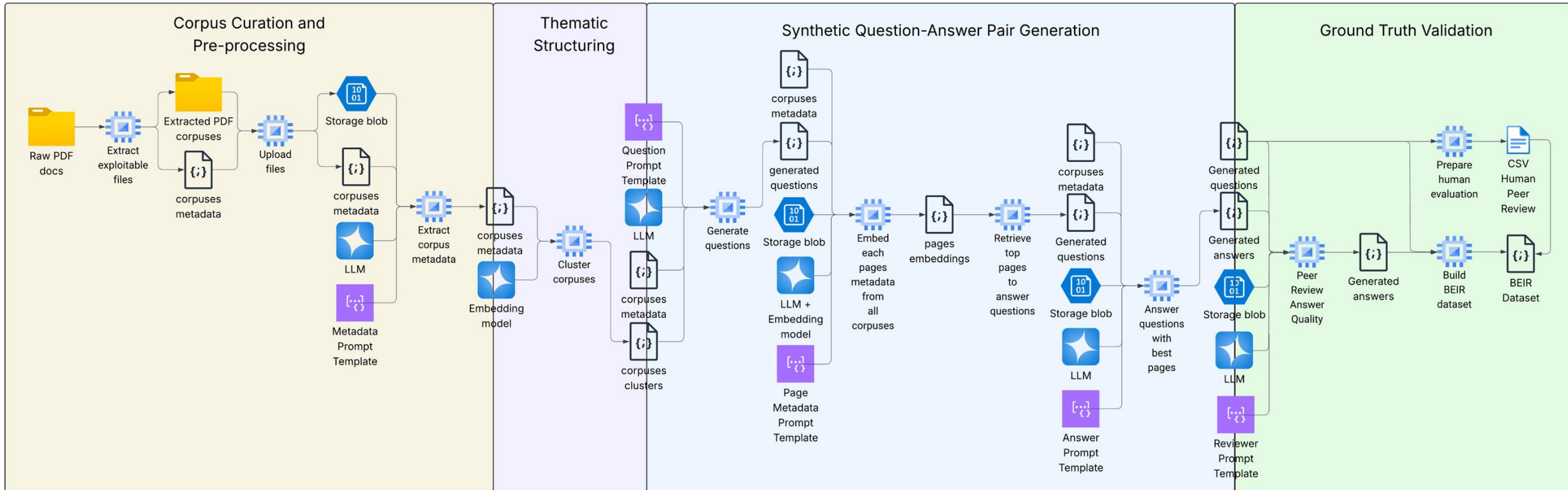
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- Lack real-world complexity
- Saturated by state-of-the-art models
- Fail to test the "Needle-in-the-Haystack" problem
- Expensive to produce

LLM-Driven Methodology

A four-stage, semi-automated pipeline to generate robust, domain-specific evaluation datasets: corpus curation; thematic structuring; synthetic QA generation; LLM Peer Review.

DATASET GENERATION



Corpus Curation

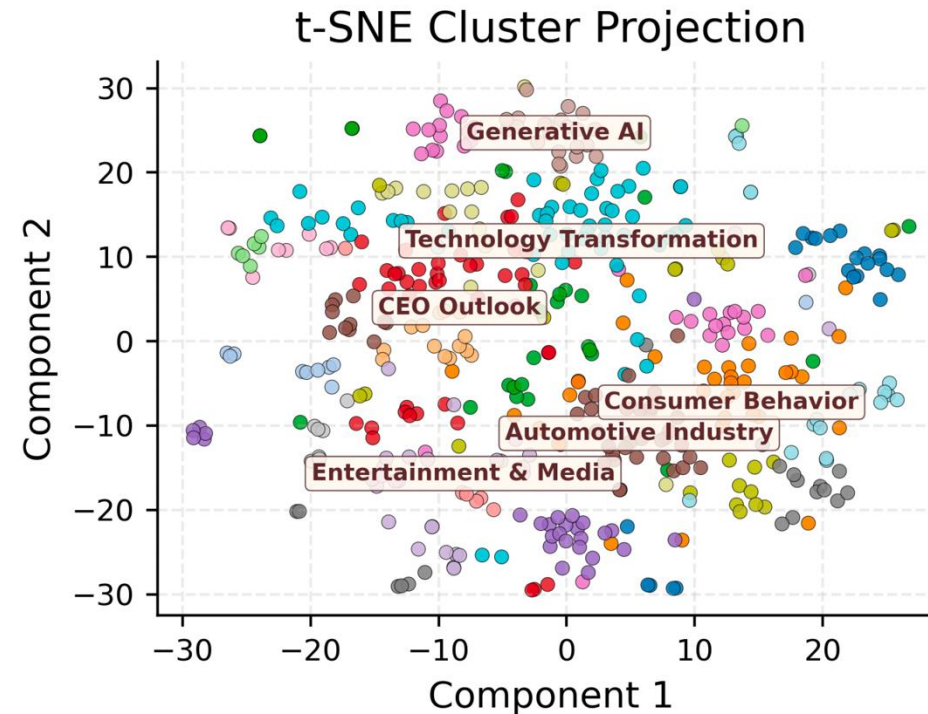
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- Cluster embeddings via K-Means



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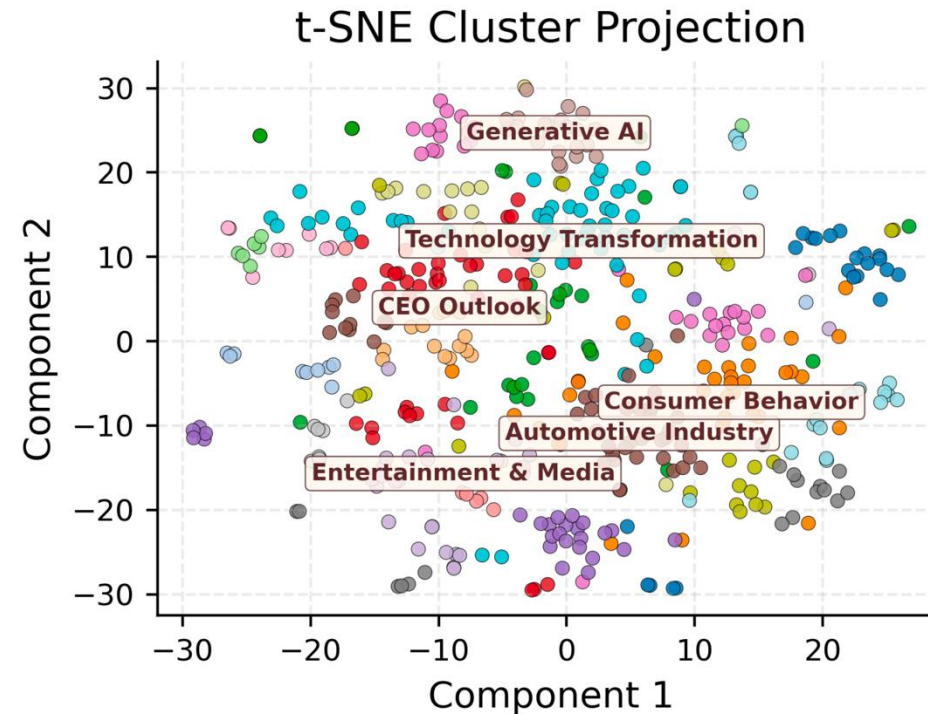
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Why clustering?

- Ease question coherence
- Avoid double questions



Synthetic QA Generation

- Recency-Bias
- Single-Fact
- Single-Slide
- Multi-Slide
- Aggregation
- Top-Level Strategic

Use Top K pages to respond

GPT (text-embedding-3-small) model to embed each page, common cosine similarity computed on all pages to extract best pages to generate the answer.

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- GPT-4o used to review the answer given the questions and relevant corpuses.
- Assess Answerability, Faithfulness and Relevance.

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Outcome

- 300 unique questions
- 1200 question-answer pairs
- Sparsity of 0.07%
- Produced BEIR format

Consulting Dataset QA sample

query-id: 162

query: How are luxury brand strategies evolving to cater to millennial and Gen Z consumers across different global markets while integrating technological and sustainability trends?

corpus:



answer:

Evolving Luxury Brand Strategies for Millennials and Gen Z

Overview and Market Trends

- ① Millennials represent 30% of the luxury market, with Gen Z emerging as influential consumers. Together, they drive approximately 85% of market growth, requiring brands to enhance value propositions.
- ② Gen Z consumers emphasize identity and experiential purchases, while millennials lead with clear preferences that influence broader market trends.

Strategies for Global Markets

- ③ **Personalization and Identity:** Brands focus on promoting products as identity statements, emphasizing "extravagant & fun" features along with traditional values like superior quality and craftsmanship.
- ④ **Technology Integration:** The digital ecosystem plays a critical role, supporting a holistic redesign across luxury value chains. This includes omnichannel strategies and influencer collaborations.
- ⑤ **Sustainability and Responsibility:** Key values include sustainability, social responsibility, and circular fashion, leading to increased preference for access over ownership and expanding secondhand markets.

Regional Adaptation

- ⑥ **Chinese Market Focus:** Strategies involve unique content on preferred sales channels, with emphasis on where and how to reach Chinese consumers.
- ⑦ **Western Markets:** Collaborations and mix-and-match approaches are pivotal, aiming to strengthen brand loyalty and extend reach through social media.

Implications and Next Steps

- ⑧ **Strategic Collaboration:** Brands should develop partnerships that elevate engagement with younger consumers through technology and personalization.
- ⑨ **Customer Experience:** Redefine customer journeys by focusing on ecosystems rather than isolated product offerings.
- ⑩ **Risks:** Balancing traditional and modern values might challenge brand identity coherence. Brands need strategic alignment with technological advancement and sustainability principles.

Hypotheses

- **H1:** MultiRAG will perform at least as well as the best single expert, showcasing robustness.
- **H2:** The effectiveness of fusion algorithms will vary by dataset, with rank-based methods being more robust to score inconsistencies.
- **H3:** Our "Consulting" dataset will be a challenging benchmark that highlights MultiRAG's advantages in real-world scenarios.

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- **H3:** Our "Consulting" dataset will be a challenging benchmark that highlights MultiRAG's advantages in real-world scenarios.

Evaluation

Benchmarks: Tested on 11 diverse datasets:

- BEIR (text-heavy)
- ViDoRe (visually-rich)
- Consulting dataset

Metrics: Focused on retrieval quality mainly:

- nDCG@5
- Recall@5
- Precision@5
- MAP@5
- MRR@5

k available: 1, 3, 10, 20, 50, 100

MultiRAG Outperforms Baselines (Validates H1)

RAG system	nDCG@5	Recall@5	Precision@5	MAP@5	MRR@5
Traditional	48.9%	42.4%	26.4%	35.9%	55.0%
Multimodal	66.2%	58.3%	32.3%	51.4%	73.8%
Jina Text Standalone	46.9%	41.9%	24.6%	34.9%	52.0%
Jina Image Standalone	57.5%	52.3%	28.1%	44.6%	65.9%
Traditional (Jina Reranker)	58.7%	50.2%	31.4%	43.5%	63.3%
Multimodal (Jina Reranker)	68.4%	60.1%	33.6%	53.0%	75.7%
Traditional (Gemini Reranker)	37.2%	34.9%	18.4%	27.3%	54.5%
Multimodal (Gemini Reranker)	57.6%	53.0%	26.0%	44.9%	73.8%
MultiRAG (rrf k=100)	61.6%	52.9%	32.2%	46.1%	68.7%
MultiRAG (rrf k=30)	61.9%	53.4%	32.3%	46.4%	68.7%
MultiRAG (L1 norm-avg)	61.9%	53.2%	32.2%	46.7%	69.0%
MultiRAG (L2 norm-avg)	62.1%	53.2%	32.3%	46.8%	69.2%
MultiRAG (L1 norm-rank)	65.2%	57.4%	33.0%	49.8%	72.3%
MultiRAG (L2 norm-rank)	65.8%	57.9%	33.2%	50.3%	72.9%
MultiRAG (max-rank)	68.6%	60.5%	34.0%	53.2%	75.5%

Across all 10 standard benchmarks, the MultiRAG (max-rank) configuration achieved the highest average scores (e.g., nDCG@5 of 68.6%).

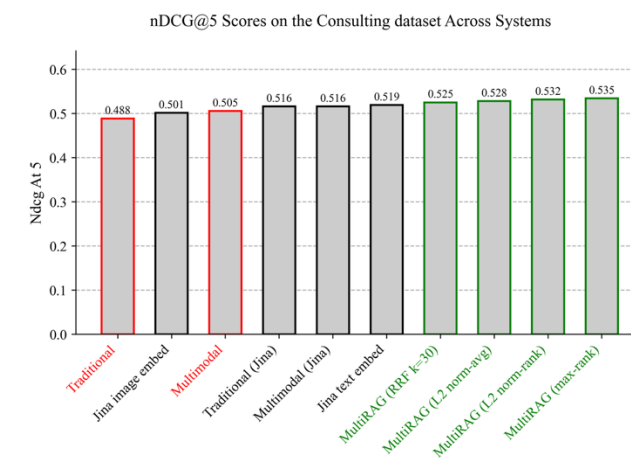
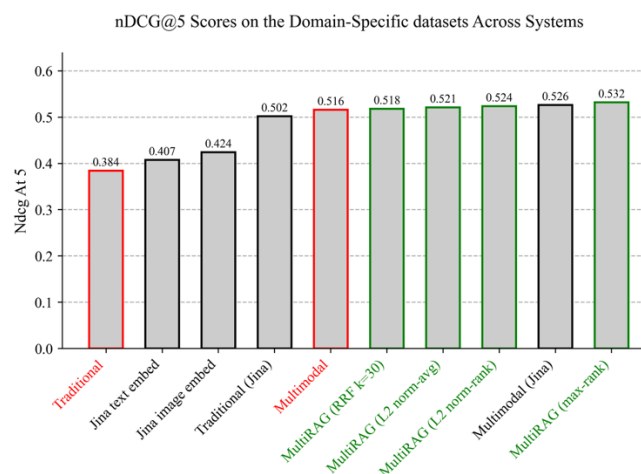
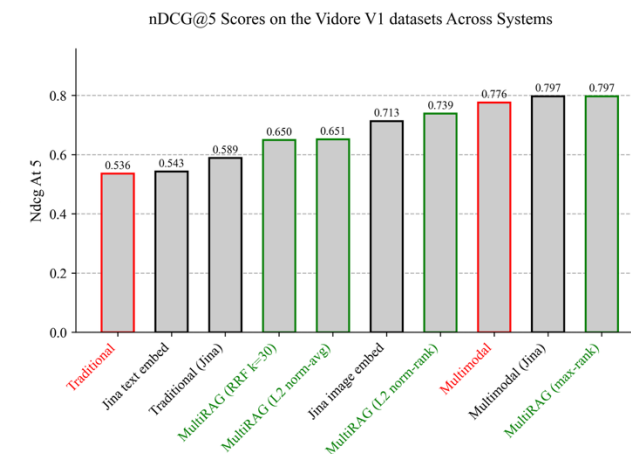
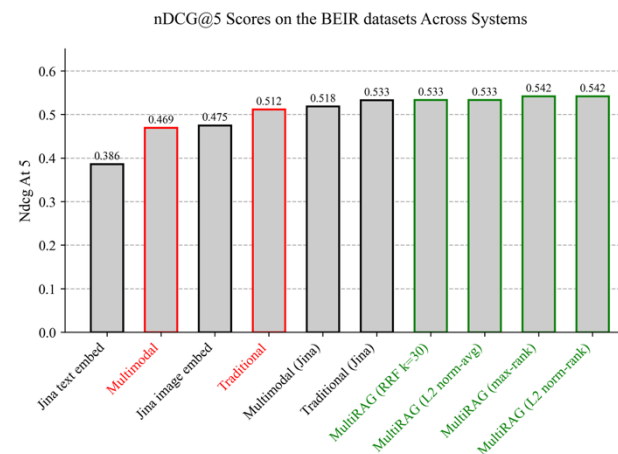
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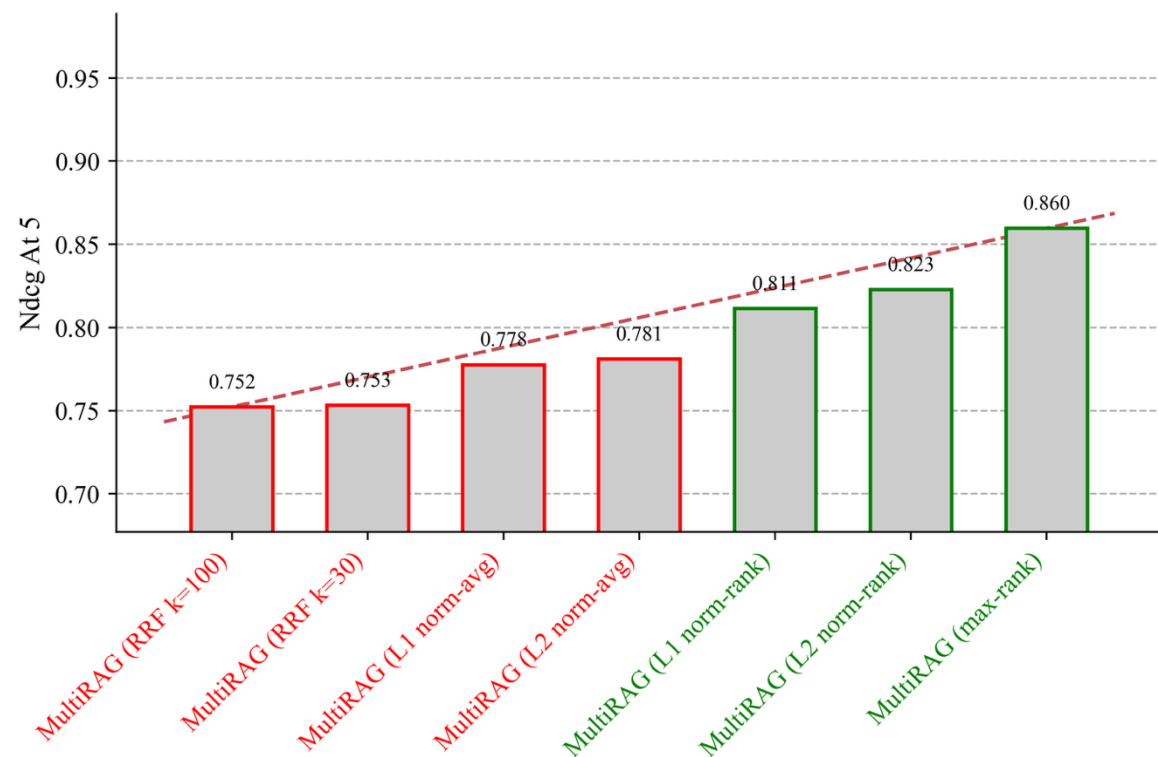
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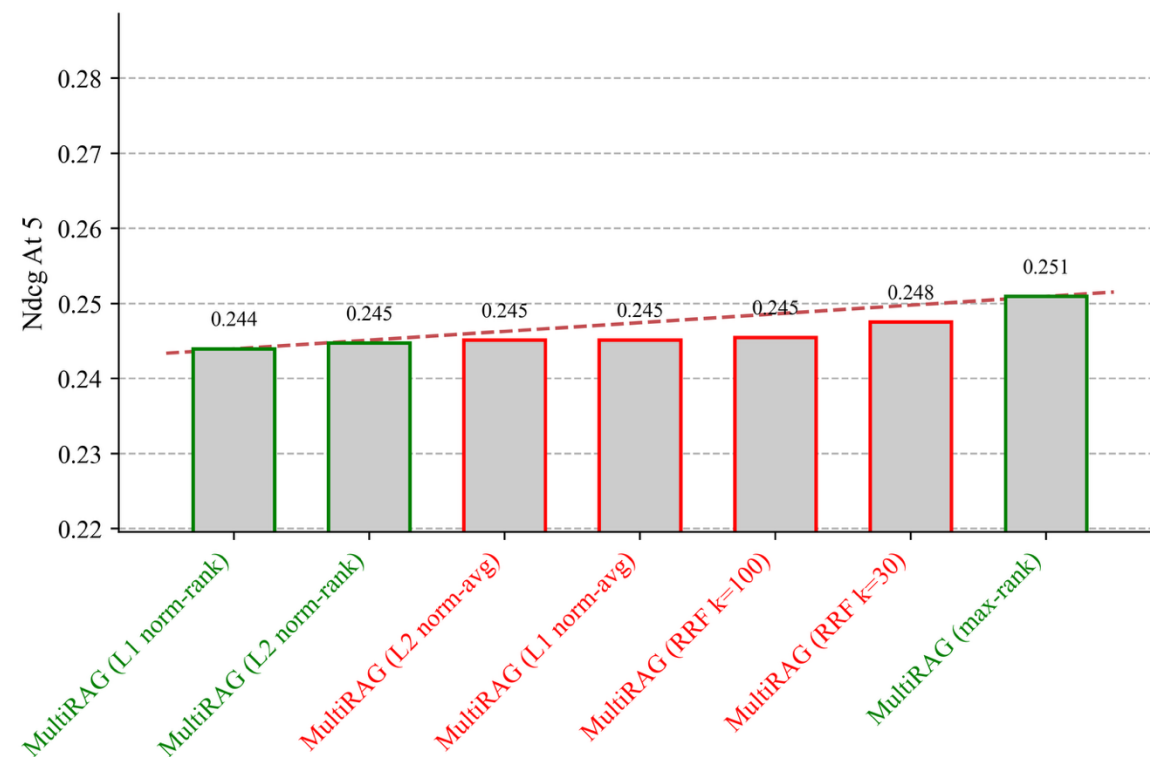
Fusion Strategy Matters (Validates H2):

The max-rank fusion strategy, which favors the strongest signal from a single expert, consistently performed best.

nDCG Scores on the InfoVQA dataset with K=5 Across Systems



nDCG Scores on the Consulting dataset with K=5 Across Systems



"Consulting" Dataset Highlights Real-World Challenges (Validates H3):

All systems showed lower performance on the "Consulting" dataset, confirming its difficulty and realism.

RAG system	Esg Reports	Economics Reports	Biomedical Lectures	Consulting Light	Consulting
Traditional	24.4%	25.2%	23.9%	37.4%	21.1%
Multimodal	53.7%	49%	56.2%	39.1%	22%
Traditional (Jina Reranker)	44.8%	50.2%	49.8%	39.3%	24.1%
Multimodal (Jina Reranker)	46.8%	56.3%	58.7%	39.9%	23%
MultiRAG (rrf k=30)	46.3%	54%	52.2%	39.7%	24.8%
MultiRAG (L2 norm-avg)	48.8%	52.6%	52%	40.9%	24.5%
MultiRAG (L2 norm-rank)	47.6%	51%	55.2%	41.3%	24.5%
MultiRAG (L2 max-rank)	48.5%	51.8%	58.3%	41.4%	25.1%

"Consulting" Dataset Highlights Real-World Challenges (Validates H3):

There was a significant performance drop (~40%) when moving from document-level to page-level retrieval, illustrating the severity of the "Needle-in-the-Haystack" problem in a large corpus.

RAG system	Consulting Light V1	Consulting Light V2	Consulting V1	Consulting V2
Traditional	76.1%	41.9%	64.9%	24.5%
Multimodal	77.1%	43.2%	67.4%	25.5%
Traditional (Jina Reranker)	79.1%	44.0%	69.0%	27.9%
Multimodal (Jina Reranker)	79.5%	44.3%	68.0%	27.2%
MultiRAG (rrf k=30)	78.9%	44.5%	70.2%	29.0%
MultiRAG (L2 norm-avg)	80.6%	45.3%	68.7%	28.5%
MultiRAG (L2 norm-rank)	81.9%	46.2%	69.6%	29.1%
MultiRAG (L2 max-rank)	81.7%	46.4%	70.2%	29.2%

A "Mixture-of-Experts" approach with fusion is more robust and effective than a single, monolithic retriever for heterogeneous data.

Limitations

- **Computational Overhead:** The framework requires maintaining multiple indexes and running parallel retrieval processes, increasing storage, compute and latency.
- **Dataset Generation Bias:** The quality of the generated dataset is inherently tied to the capabilities and potential biases of the LLM used in the pipeline.

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Future Work

- **Expand the Expert Ensemble:** Incorporate other experts like Graph RAG for relational reasoning or SQL retrievers for structured data.
- **Agentic Fusion:** Develop a dynamic fusion layer where an LLM agent predicts the best expert(s) to use for a given query.
- **Apply Dataset Pipeline to New Domains:** Use the generation methodology to create new benchmarks for fields like law, finance, or biomedical research.

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Thank you